

DISCRETE SPECTRUM GIVES PREVALENT CYCLIC VECTORS

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SOURCE PROBLEM

The source paper is Koltai–Kunde, *A Koopman–Takens theorem: Linear least squares prediction of nonlinear time series*, arXiv:2308.02175v3. On page 17 the authors formulate the following conjecture.

Conjecture 25. Let the ergodic $T \in \text{MPT}$ possess discrete spectrum. Then $\mathcal{U} = \mathcal{U}_T$ has a prevalent set of cyclic vectors in L_μ^2 .

It is natural to ask whether for an ergodic transformation the possession of discrete spectrum already implies that the set of cyclic vectors is prevalent. The Representation Theorem [Wal00, Theorem 3.6] states that an ergodic measure-preserving transformation with discrete spectrum on a probability space is isomorphic to an ergodic rotation on a compact abelian group. The strong analogy between (ergodic) circle rotations and group rotations, together with the prevalence statement of Proposition 19 could lead one to formulate the following conjecture:

Conjecture 25. Let the ergodic $T \in \text{MPT}$ possess discrete spectrum. Then $\mathcal{U} = \mathcal{U}_T$ has a prevalent set of cyclic vectors in L_μ^2 .

In Section 3.5 we will discuss another type of systems that possesses cyclic vectors.

3.4. A generic MPT has a dense G_δ set of cyclic vectors

By Theorem 23 the collection of measure-preserving transformations with a dense G_δ set of cyclic vectors lies dense in MPT with respect to the weak topology. In fact, we can

RESULT

Theorem 1. Let T be an ergodic invertible measure-preserving transformation on the probability space considered in the source paper, and let $\mathcal{U} = \mathcal{U}_T$ act on the complex Hilbert space $H = L_\mu^2$. If T has discrete spectrum, then the set of cyclic vectors for $\mathcal{U}$ is prevalent. In fact, after writing H as an orthogonal sum of one-dimensional eigenspaces,

$$H = \overline{\text{span}}\{e_j : j \in J\}, \quad \mathcal{U}e_j = \lambda_j e_j,$$

a vector $f = \sum_{j \in J} a_j e_j$ is cyclic if and only if $a_j \neq 0$ for every $j \in J$. Consequently the cyclic vectors also contain a dense G_δ set.

IDEA OF THE PROOF

Discrete spectrum diagonalizes the Koopman operator. Ergodicity rules out eigenvalue multiplicity: two eigenfunctions with the same eigenvalue have an invariant quotient, hence are scalar multiples. Thus the problem becomes the cyclic-vector problem for a diagonal unitary with distinct atoms on the unit circle.

For a vector with all coordinates nonzero, the scalar spectral measure is purely atomic and charges every eigenvalue. Kolmogorov's density theorem, quoted in the source paper, says analytic polynomials are dense in L^2 of any singular measure on $\mathbb{T}$. Therefore positive Koopman powers of the vector recover every eigendirection. Finally, the set of vectors with no zero coordinate is prevalent because a single probe vector with no zero coordinate misses all coordinate hyperplanes except at countably many probe parameters.

PROOF

Since T has discrete spectrum in the sense of the source paper, $H = L^2_\mu$ has a countable orthonormal basis of eigenfunctions for $\mathcal{U}$. We first note that each eigenspace is one-dimensional. Indeed, if $\mathcal{U}g = \lambda g$, then $|g| \circ T = |g|$, so ergodicity implies that $|g|$ is almost everywhere constant. Thus a nonzero eigenfunction is nonzero almost everywhere after changing it on a null set. If g and h are eigenfunctions for the same eigenvalue λ , then h/g is T -invariant almost everywhere, hence constant by ergodicity. Thus $\text{span}\{g, h\}$ is one-dimensional.

Index the eigenlines once, by a countable set J , and choose normalized eigenvectors e_j with pairwise distinct eigenvalues $\lambda_j \in \mathbb{T}$. Let

$$f = \sum_{j \in J} a_j e_j \in H.$$

If $a_k = 0$ for some k , then every vector in $\text{span}\{\mathcal{U}^n f : n \geq 0\}$ has zero e_k -coordinate. Hence the cyclic subspace cannot be all of H , and f is not cyclic.

Conversely assume $a_j \neq 0$ for all j . Let

$$\nu_f = \sum_{j \in J} |a_j|^2 \delta_{\lambda_j}$$

be the scalar spectral measure associated to f , up to the harmless normalizing factor $\|f\|^{-2}$ if one wants a probability measure. Under the spectral transform for the cyclic subspace generated by f , the vector $p(\mathcal{U})f$ corresponds to the analytic polynomial p in $L^2(\nu_f)$. The measure ν_f is purely atomic and therefore singular with respect to Haar measure on $\mathbb{T}$. Kolmogorov's density theorem [2, Theorem 2.11.5], also quoted as Theorem 13 in the source paper, implies that analytic polynomials are dense in $L^2(\nu_f)$.

Thus the closure of $\{p(\mathcal{U})f : p \text{ an analytic polynomial}\}$ is the whole Hilbert space represented by $L^2(\nu_f)$. Since every atom λ_j has positive ν_f -mass, this represented space is exactly the closed span of all eigenvectors e_j , namely H . Hence f is cyclic. We have proved the characterization:

$$f \text{ is cyclic} \iff \langle f, e_j \rangle \neq 0 \text{ for every } j \in J.$$

It remains to prove prevalence. Enumerate $J = \{1, 2, \dots\}$, with the obvious finite modification if J is finite, and set

$$p = \sum_{j \geq 1} 2^{-j} e_j \in H.$$

Let $E = \mathbb{C}p$, a one-dimensional complex probe space. For an arbitrary $h = \sum b_j e_j \in H$, the j -th coordinate of $h + zp$ is

$$b_j + z 2^{-j}.$$

For each fixed j this coordinate vanishes for at most one value of $z \in \mathbb{C}$. Therefore the set of z for which at least one coordinate of $h + zp$ vanishes is countable, hence has two-dimensional Lebesgue measure zero in $E \simeq \mathbb{C}$. For almost every $z \in \mathbb{C}$, all coordinates of $h + zp$ are nonzero, so $h + zp$ is cyclic. This is exactly prevalence in the sense used in the source paper.

The same characterization also gives a dense G_δ set, since

$$\bigcap_{j \in J} \{f \in H : \langle f, e_j \rangle \neq 0\}$$

is a countable intersection of open dense sets. This proves the conjecture.

VERIFICATION AND NOVELTY NOTES

The proof uses only standard ingredients: diagonalization for discrete spectrum, simple eigenspaces for ergodic transformations, Kolmogorov’s density theorem for singular spectral measures, and the definition of prevalence via a finite-dimensional probe.

Local run indexes were searched for the arXiv id and terms including “Koopman”, “Takens”, “cyclic vectors”, “discrete spectrum”, and “prevalent”. No exact prior packet for Conjecture 25 was found. Web searches for exact phrases around the conjecture and its label did not reveal a later paper explicitly resolving it. The arXiv page records arXiv:2308.02175v3 as submitted on 2023-08-04 and last revised on 2024-03-31, with subjects math.DS and math.FA.

Novelty confidence is moderate: the result is likely valid, but the argument is short enough that it may be folklore or implicit in standard spectral theory. Human review should focus on whether the use of positive powers $\mathcal{U}^n$, $n \geq 0$, is fully justified by Kolmogorov density in the purely atomic measure.

REFERENCES

- [1] P. Koltai and P. Kunde, *A Koopman–Takens theorem: Linear least squares prediction of nonlinear time series*, arXiv:2308.02175v3, 2024.
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